



Tiles

A phone that adapts

A native HarmonyOS home of big, live tiles that reshapes itself around the person using it: seniors, kids, people with low vision, and everyone else. Ask in plain words; the assistant answers with new tiles.

ArkTS + ArkUI

API 20+ · target API 24

Human-Centric

Intelligent Experiences



Illustration in HTML/CSS, not an app screenshot

THE PROBLEM

One phone layout. Four very different people.

Phones are designed for an average user who does not exist. Settings that could help are buried, and nobody adjusts them for the people who need it most.

Senior

Halina, 78

Tiny icons, missed taps, forgotten pills. A "your grandson needs money, right now" call nearly worked.

Kid

Zosia, 8

Too many choices and endless feeds. Homework, a focus timer and "call Mum" would do.

Low vision

Michał

Small targets and silent screens. Needs fewer, larger tiles and everything read aloud.

Everyday

Daniel, 29

Plans spread over five chats. "Dinner Friday, vegan, under 80 zł" should be one tile with a vote.

Accessibility settings exist, but they are one-size switches. Nothing looks at **how this person actually uses the phone** and adapts, with family in the loop.

THE SOLUTION

A home made of live tiles, generated for you

Live, actionable tiles

Each tile is a tiny card that does one thing: **"Pills 8:00, tap when taken"**, **"Call Kasia"**, **"14° and rain, take the umbrella"**, **"Homework 3 of 5"**. Big, colourful, readable aloud.

Ask Tiles

Type a request in English or Polish. The assistant answers with **new tiles**, not paragraphs: a reminder, a call shortcut, a dinner plan with vote buttons.

Adapts, family approves

Missed taps and unused tiles become suggestions ("text 40% bigger?"). For seniors and kids, a **caregiver or parent approves** every change first.

Person asks

"Remind me to take Euthyrox at 8 every morning"



Composer

on-device intent planner: medicine, time, repeat



TileSpec

pills tile · sun · 2×1 · remind + done



Approval

senior or kid: waits for Kasia / Mum



Home

tile springs in; reminder scheduled

Plus **Guardian**, a scam shield for seniors: explainable red flags in a suspicious call, one calm instruction, family alerted.

The assistant outputs interface, not text

1 · REQUEST

Dinner Friday with Ola and Kuba,
vegan, near Kazimierz, under 80 zł

↓ slots extracted on the device

day: Friday people: Ola, Kuba food: vegan
place: Kazimierz budget: 80 zł

- EN + PL keywords: time, day, person, place, food, budget, medicine, homework subject.
- Deterministic and offline. No model, no network.

2 · TILESPEC (THE CONTRACT)

```
{ "type": "plan", "size": "2x2", "color": "peach",  
  "title": "Dinner Fri 19:30",  
  "subtitle": "Ola, Kuba · vegan · ≤ 80 zł",  
  "blocks": [  
    { "kind": "chips",  
      "items": ["Kazimierz", "vegan", "80 zł"] },  
    { "kind": "list",  
      "items": ["Veganda", "Krowarzywa"] },  
    { "kind": "buttons", "actions": [  
      { "label": "Vote Veganda", "kind": "vote" },  
      { "label": "Remind me", "kind": "remind" } ] }  
  ],  
  "source": "assistant" }
```

3 · RENDERED NATIVELY (ARKUI)

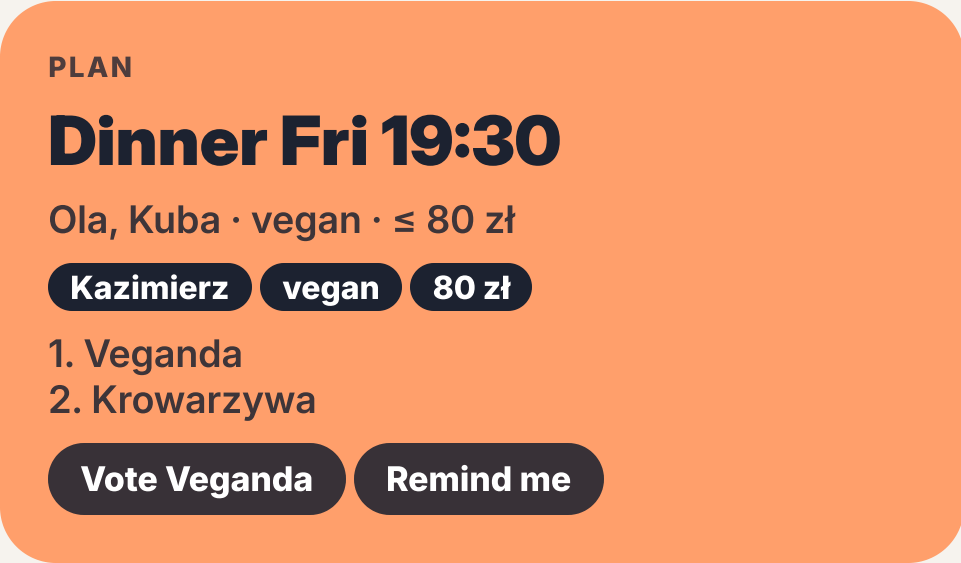


Illustration of the rendered tile

- One renderer for every tile: text, big, list, checklist, progress, timer, chips, alert, weather, buttons.

The generator is pluggable: the same TileSpec schema could later come from a language model behind local-first guardrails. Today it is a rule-based planner, and the app says so.

The same app regenerates itself for each person

Halina, 78

2 columns, text ×1.4, high contrast, read-aloud, caregiver Kasia

PILLS 8:00

Tap when taken

CALL

Kasia

GUARDIAN

Scam shield

SOS

Hold for help

Zosia, 8

Playful palette; anything new waits for Mum

HOMEWORK

3 of 5 done

FOCUS

20 min

CALL

Mum

SCREEN TIME

45 min left

TODAY'S ADVENTURE

Find 3 red leaves

Michał

Voice-first, fewer and larger tiles, everything spoken

SAY

"Read my tiles"

BUS 152

in 6 min

CALL

Brother

Daniel, 29

Dense, colourful grid: plans, focus, commute

PLAN

Dinner Fri

TRAM

8 min

FOCUS

25:00

GYM

18:00

WEATHER

14° rain

A profile (`Profile`: text scale, high contrast, read-aloud, columns, caregiver) drives the theme, the grid and the seed tiles. Illustrations; real emulator screenshots on slide 9.

Learns from use. Family stays in the loop.

Adaptation, approved by the caregiver

Usage signals (missed taps, unused tiles, time of day) become plain suggestions. Kasia approves them on the Caregiver screen, shown in the demo as a "Kasia's phone" simulation.

11 missed taps today

Make text 40% bigger and read tiles aloud?

Approve

Decline

New tile from the assistant

"Euthyrox 8:00 every morning" waits for Kasia

Guardian scam shield

For seniors: a suspicious call or SMS is checked for explainable red flags, then the screen becomes one calm instruction.

asks for money

pressure: right now

"don't tell anyone"

**This looks like a scam.
Hang up. Kasia already knows.**

Demo input is simulated and labelled "**Demo call**". Checks run on the phone.

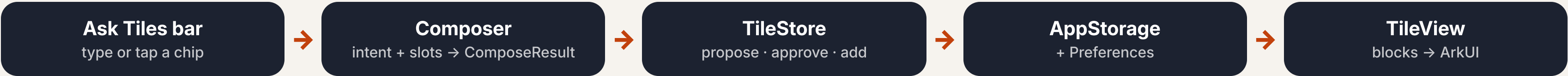
USE OF PLATFORM CAPABILITIES

Native HarmonyOS, not a web view

CAPABILITY	KIT / API	STATUS
Notifications	Notification Kit <code>publish</code> , permission dialog; <code>wantAgent</code> tap opens the tile	verified on emulator
Home-screen widgets	Form Kit: Next up 2×2, My tiles 2×4, Big buttons 4×4; <code>updateForm</code> , <code>postCardAction</code>	verified on emulator
Call a person	Telephony Kit <code>call.makeCall</code> dial screen; <code>Want</code> fallback	verified on emulator
Guardian scam shield	On-device EN/PL red-flag rules, redacted caregiver alert	verified on emulator
Generative UI assistant	On-device intent planner → <code>TileSpec</code> → native renderer	verified on emulator
Persistence + theming	ArkData Preferences; text scale, high contrast, dark mode, springs	verified on emulator
Reminders	<code>reminderAgentManager</code> → 1700002 without AppGallery entitlement; in-app timer + notification while open	fallback only
Read-aloud	Core Speech Kit not in the OpenHarmony SDK → screen-reader announcement + "Speaking..." state	via screen reader
Haptics	Sensor Service Kit <code>vibrator</code> (emulator has none)	implemented

Ordinary-app APIs only, no system privileges. Permissions: VIBRATE, PUBLISH_AGENT_REMINDER. "Verified on emulator" = seen working on the DevEco HarmonyOS 6.1.1 (API 24) emulator. Details: README "Platform capabilities used".

A schema in the middle, native UI on both sides



Model

TileSpec, TileBlock, TileAction, Profile, Suggestion, ComposeResult: one fixed contract between generator and renderer.

tiles/model/Tile.ets

Intelligence (on device)

Composer: deterministic EN/PL intent planner → 1-3 TileSpecs + reply + follow-up chips. Adaptation: usage signals → suggestions. Guardian: red-flag rules.

tiles/ai/Composer.ets tiles/ai/Adaptation.ets

State + platform

TileStore singleton (AppStorage, persisted with Preferences); approvals queue; tap recording. Platform services: widgets, notifications, reminders, read-aloud, haptics, dialer.

tiles/data/TileStore.ets tiles/platform/*

Views

Home grid, tile detail, assistant sheet, onboarding + profile picker, caregiver, Guardian, settings.

pages/Index.ets tiles/views/*

Renderer

One TileView renders any TileSpec; block components for text, list, checklist, progress, timer, chips, alert, weather, buttons.

tiles/components/*

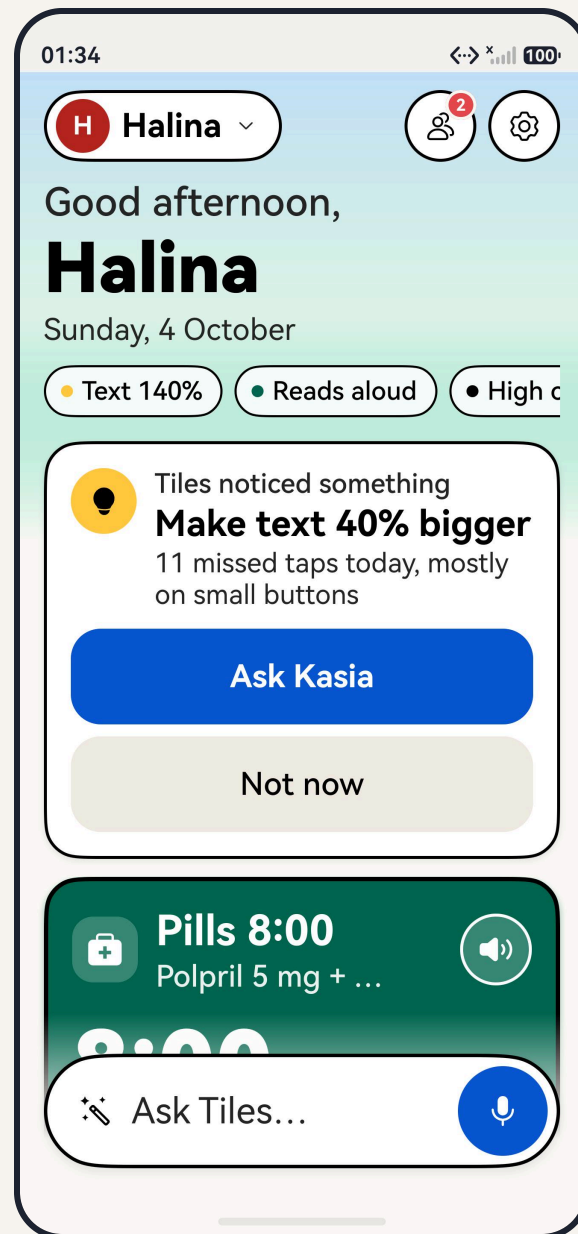
Widget

Form Kit card shows the top tiles and the next reminder, refreshed from the app.

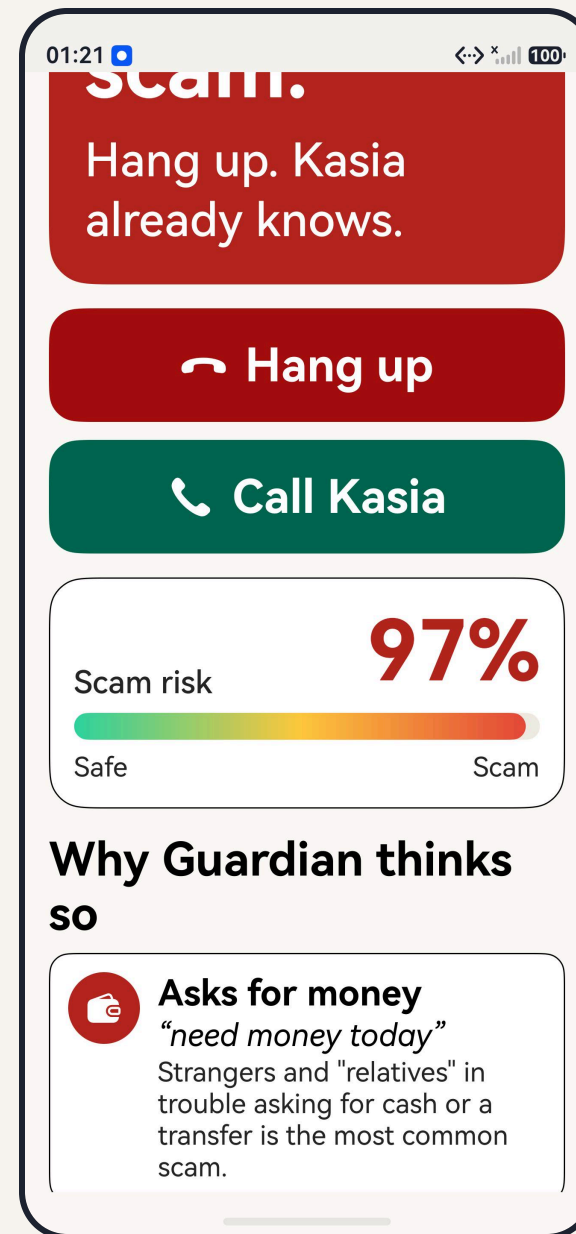
FormExtensionAbility

DEMO

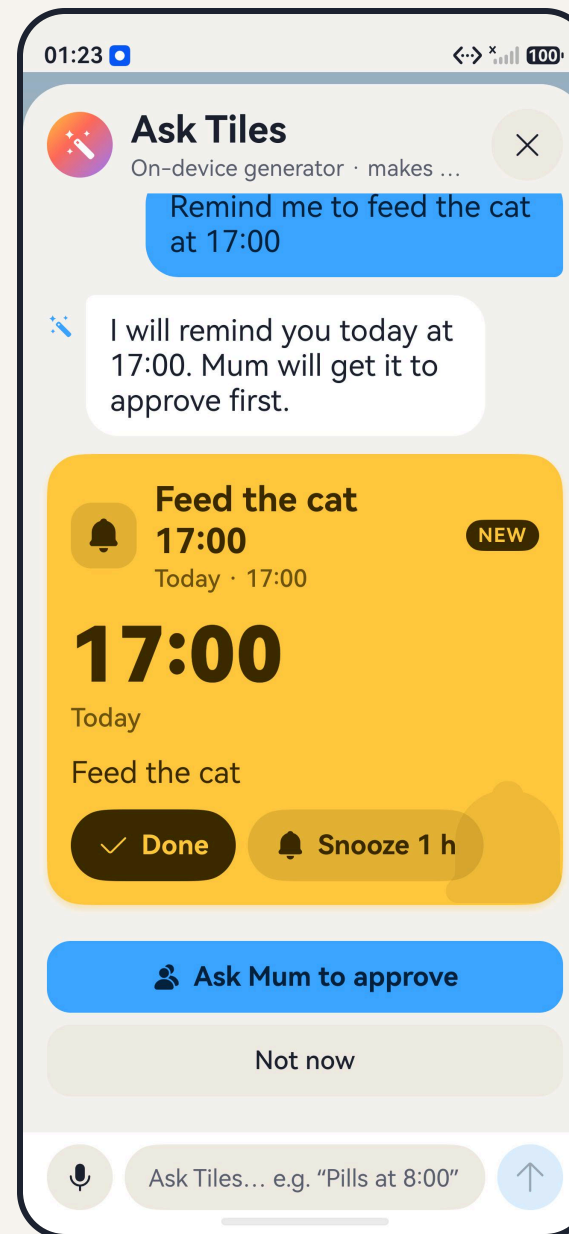
Running on the HarmonyOS 6.1 emulator



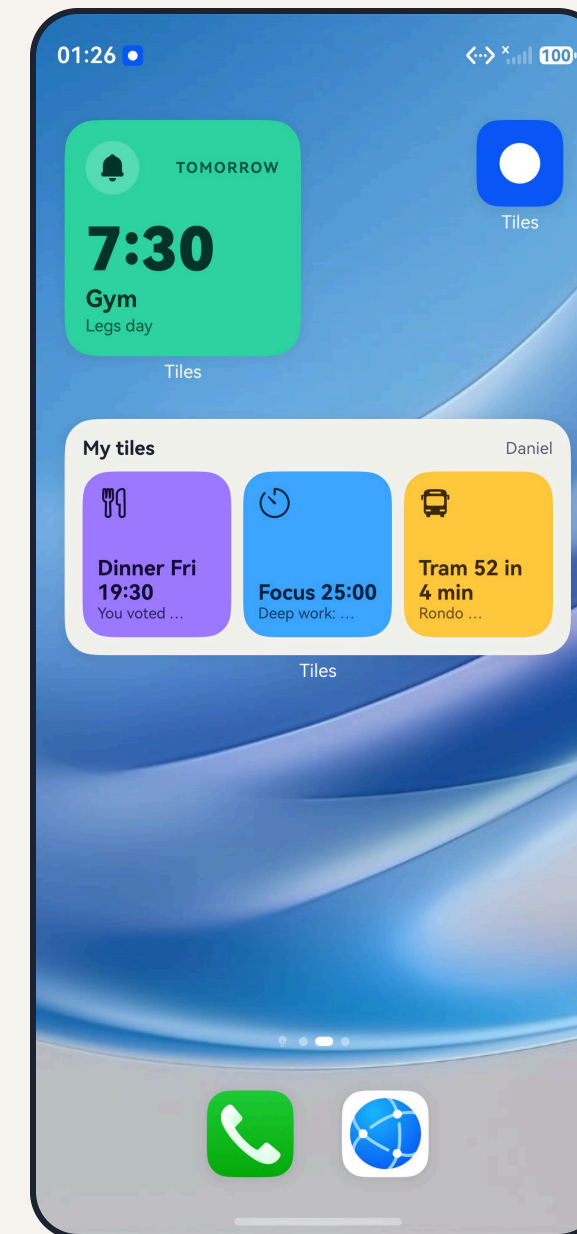
senior home tiles noticed



guardian scam risk



kid ask tiles reminder



home screen widgets

1. **Onboarding**: "Who is this phone for?" → Senior.
2. **Halina's home**: huge tiles; Pills → "Taken"; read-aloud.
3. **Ask Tiles**: "Remind me to take Euthyrox at 8" → a tile is born → waits for Kasia.
4. **Caregiver** (simulated "Kasia's phone"): approve the tile and "text 40% bigger".
5. **Guardian**: labelled demo call → red flags → "Hang up. Kasia already knows."
6. **Zosia** and **Daniel**: the home regenerates; dinner plan tile with votes.
7. **Widget** with the next reminder on the home screen.

All people and data are fictional. The assistant is an on-device rule-based generator; the Guardian call is simulated.

From a clean checkout to a running app

```
git clone https://github.com/JustAnotherDevv/tiles-hackyeah-2026
cd tiles-hackyeah-2026
```

```
# no Huawei ID: OpenHarmony SDK + its debug keystore
source .toolchain/env.sh      # README section 4.1
scripts/build-ohos-signed.sh # → release/*.hap
```

```
hdc install -r release/<signed>.hap
hdc shell aa start -a EntryAbility -b com.hackyeah.tiles
```

- **Minimum API 20**, compiled against API 24; ordinary-app permissions only.
- **Honest AI**: the assistant is a deterministic on-device planner, not an LLM. Development used Claude Code; see `AI_WORKFLOW.md`.
- **Simulated and labelled**: Guardian demo call, "Kasia's phone", weather and plan options. Reminders fire only while the app runs (no AppGallery entitlement).
- **Next**: an LLM generator behind local guardrails, a real caregiver device via distributed data, call screening on a device.

REPOSITORY

`github.com/JustAnotherDevv/tiles-hackyeah-2026`

TEAM

JustAnotherDevv